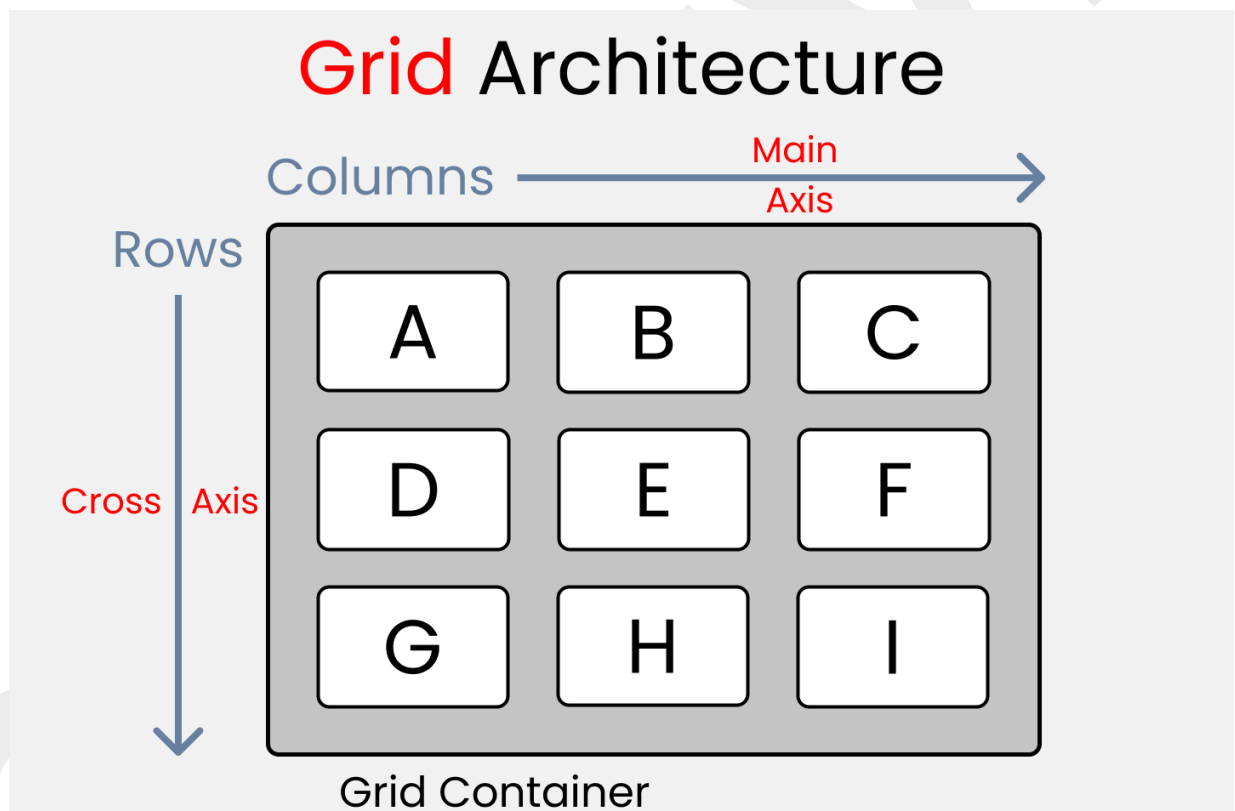


CSS Grid Properties

Grid Architecture

- So how does Grid architecture work? The Grid items [Contents] are distributed along the main axis and cross axis. Using various Grid properties, you can manipulate the items to create your website layouts.



- By the way, you can join multiple rows and columns, just like in Excel software, which gives you more flexibility and options than Flexbox.

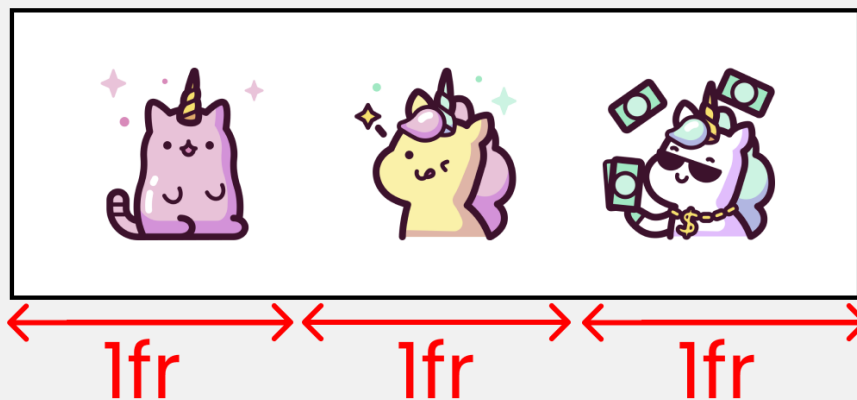
The grid-template-columns property:

- You use this property to define the number and width of columns. You can either individually set the width of each column, or set a uniform width for all columns using the repeat() function.

grid-template-columns : 200px auto 100px;



grid-template-columns : repeat(3, 1fr);



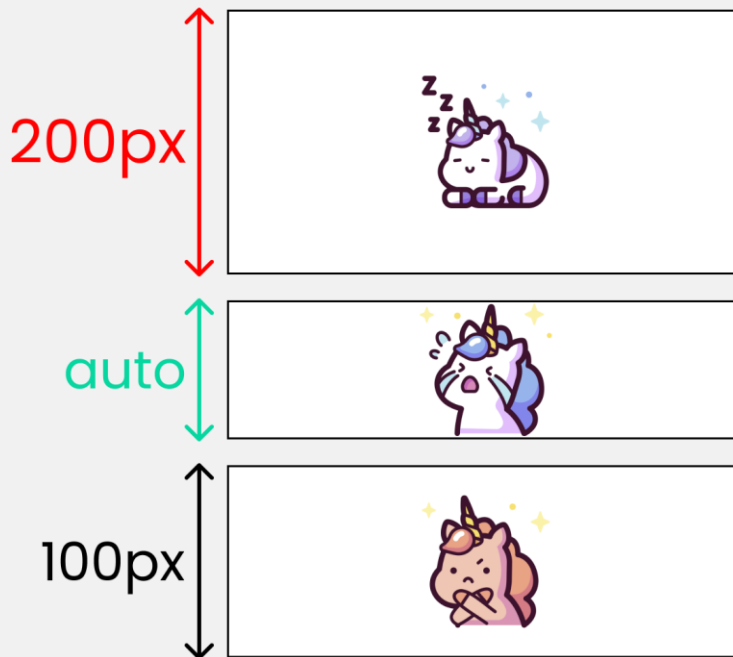
- **Note:**

- The pixel values will be an exact measurement. The "auto" keyword will cover the available space.
- If you use fr (fraction unit) as a unit of measurement, all the boxes will be equal in size.

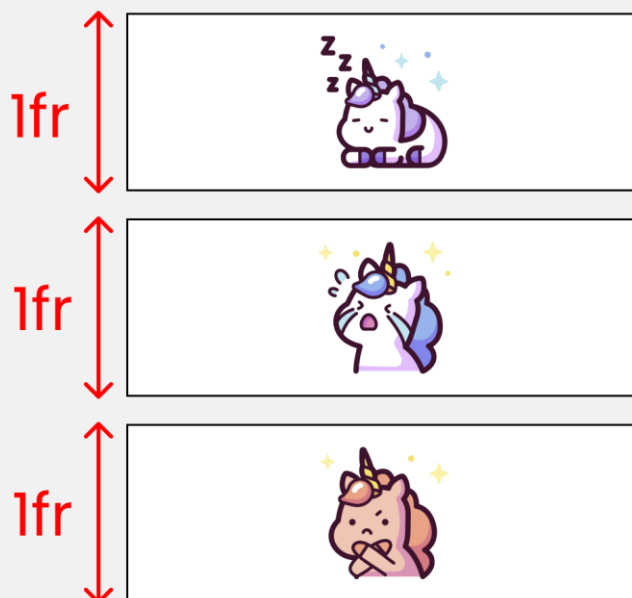
The grid-template-rows property:

- You use this property to define the number and height of rows. You can either individually set the height of each row, or set a uniform height for all rows using the repeat() function.

grid-template-rows : 200px auto 100px

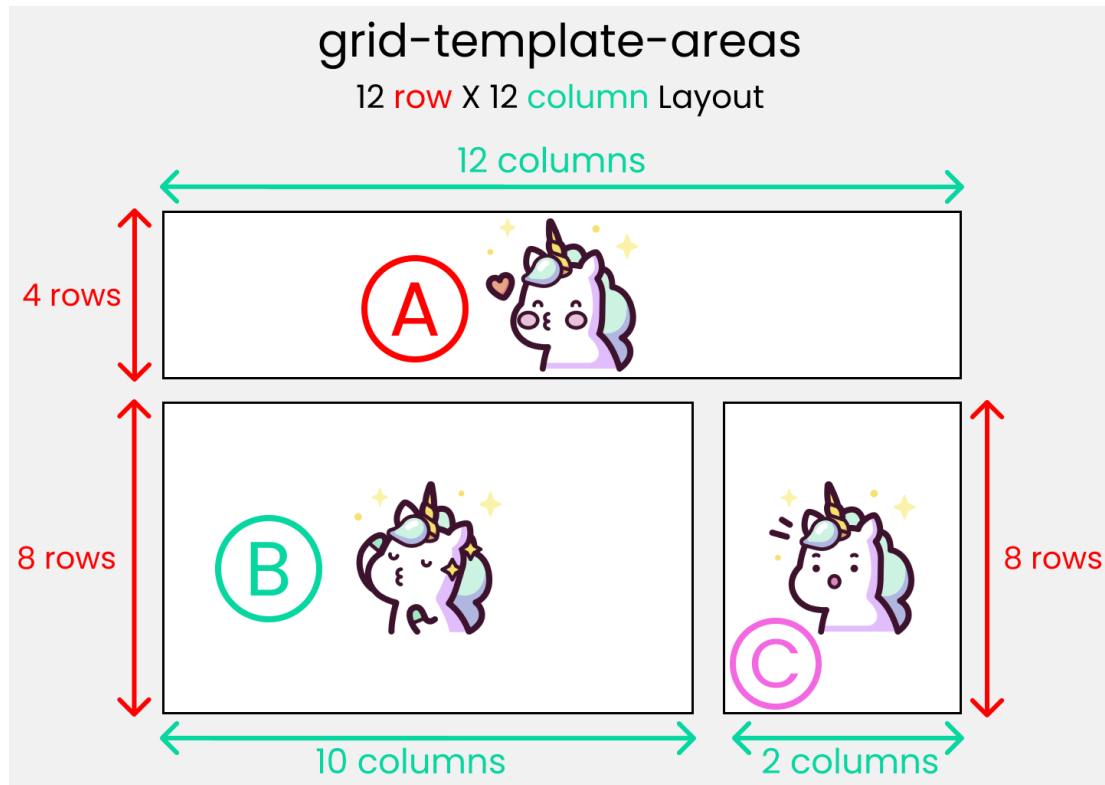


grid-template-rows : repeat(3, 1fr);



✚ The grid-template-areas property :

- You use this property to specify the amount of space a grid cell should carry in terms of columns and rows across the parent container. Life's much easier with this property as it allows us to see visually what we're doing.



The Code

grid-template-areas :

"A A A A A A A A A A A A A A"

"B B B B B B B B B B B B C C"

"B B B B B B B B B B B B C C";

- To experiment with this, you need to understand both the parent and child properties:
 - **grid-template-areas**: The parent property that will create the blueprint
 - **grid-area**: the child property that will follow the blueprint.

✚ The Column-gap Property :

- You use this property to place a gap between **Columns** inside the grid

column-gap: 50px



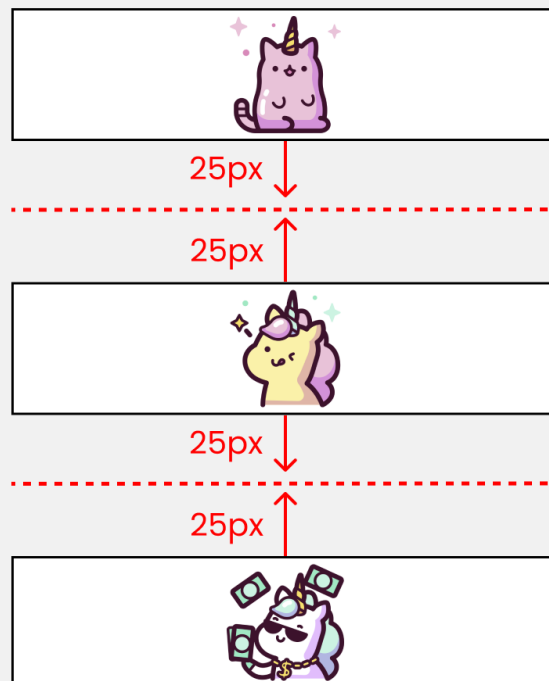
Red Dotted lines are called -> grid lines

Note: column-gap works with grid-template-columns.

✚ The row-gap Property :

- You use this property to place a gap between **Rows** inside the grid

row-gap: 50px



Red Dotted lines are called -> grid lines

The justify-items property:

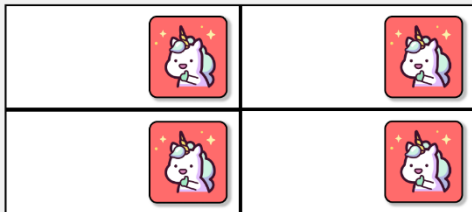
- You use this property to position grid-items (children) inside grid containers along the X-Axis [Main Axis]. The 4 values are 

justify-items -> X Axis

Start



end



justify-items -> X Axis

center



stretch



The align-items property:

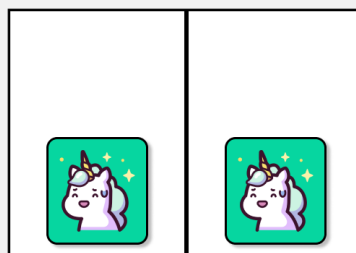
- You use this property to position grid-items (children) inside the grid container along the Y-Axis [Cross Axis]. The 4 values are 

align-items -> Y Axis

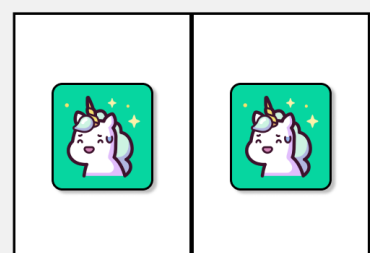
Start



end



center

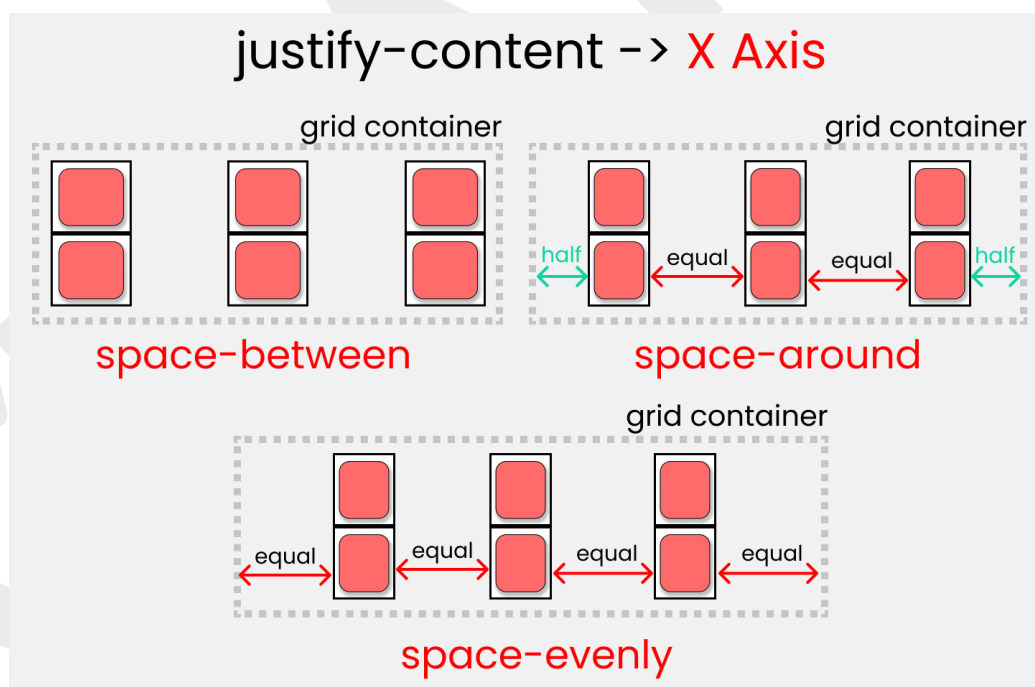
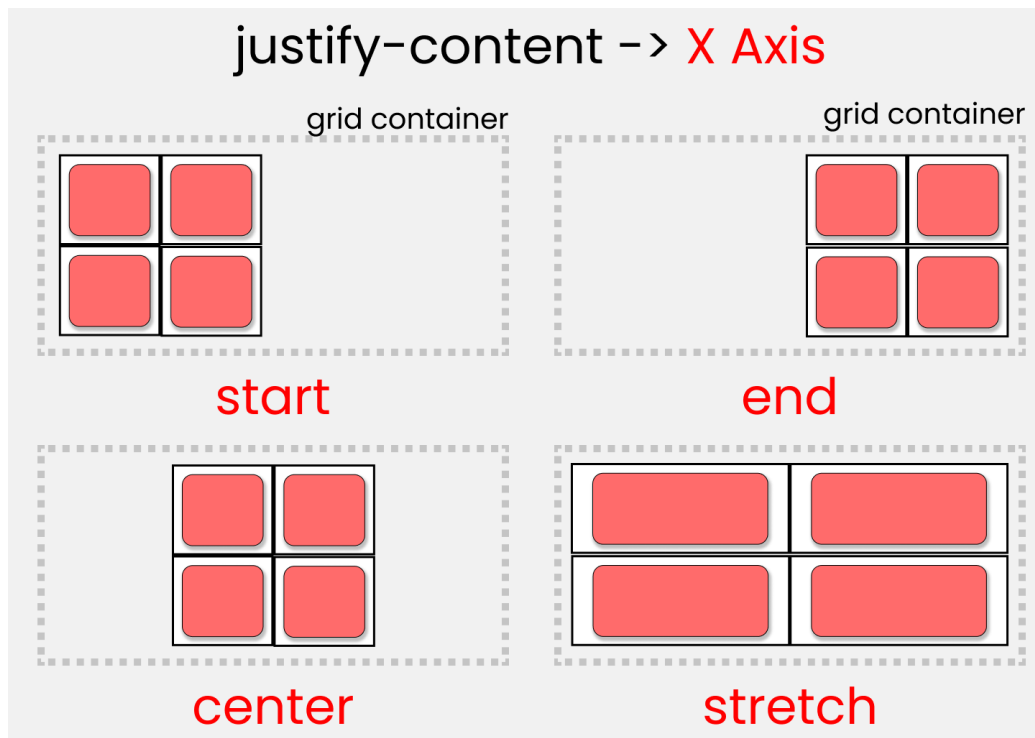


stretch



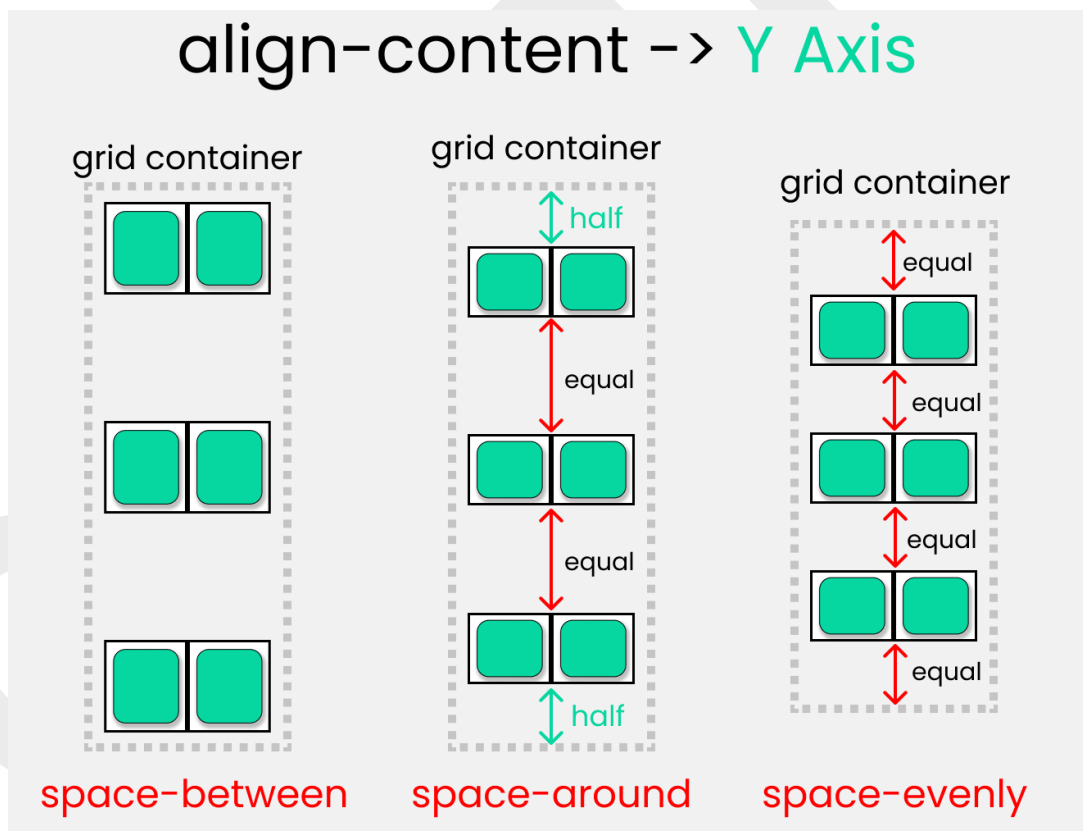
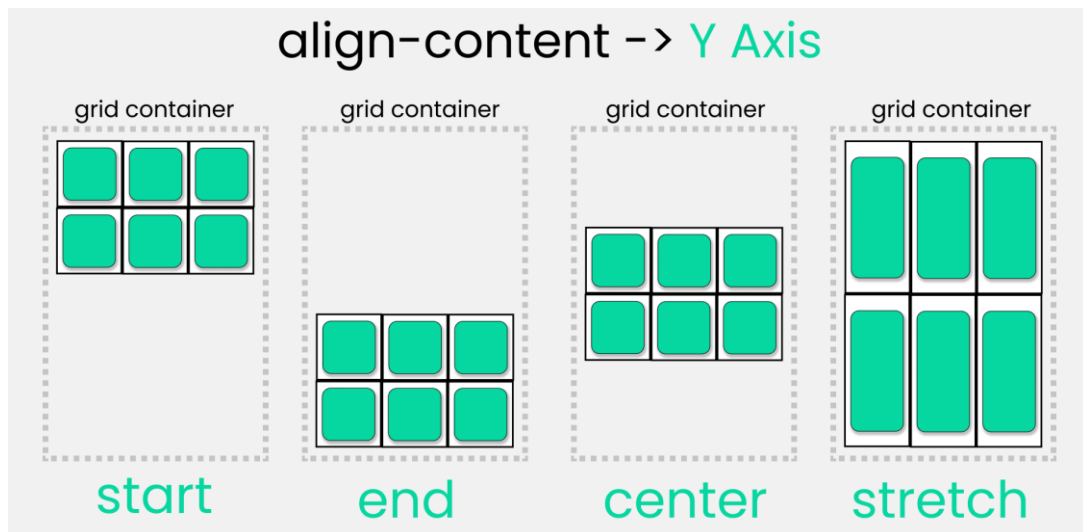
The justify-content property:

- You use this property to position your grid [Basically everything] inside the grid container along the X-Axis [Main Axis]. The 7 values are 



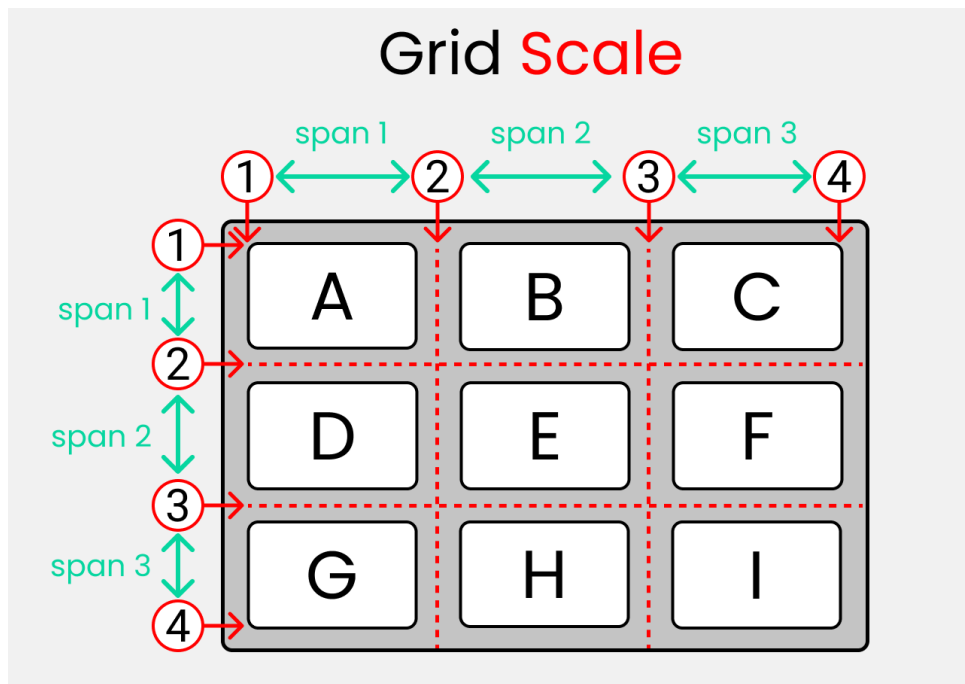
The align-content property:

- You use this property to position our grid [Basically everything] inside the grid container along the Y-Axis [Cross Axis]. The 7 values are 

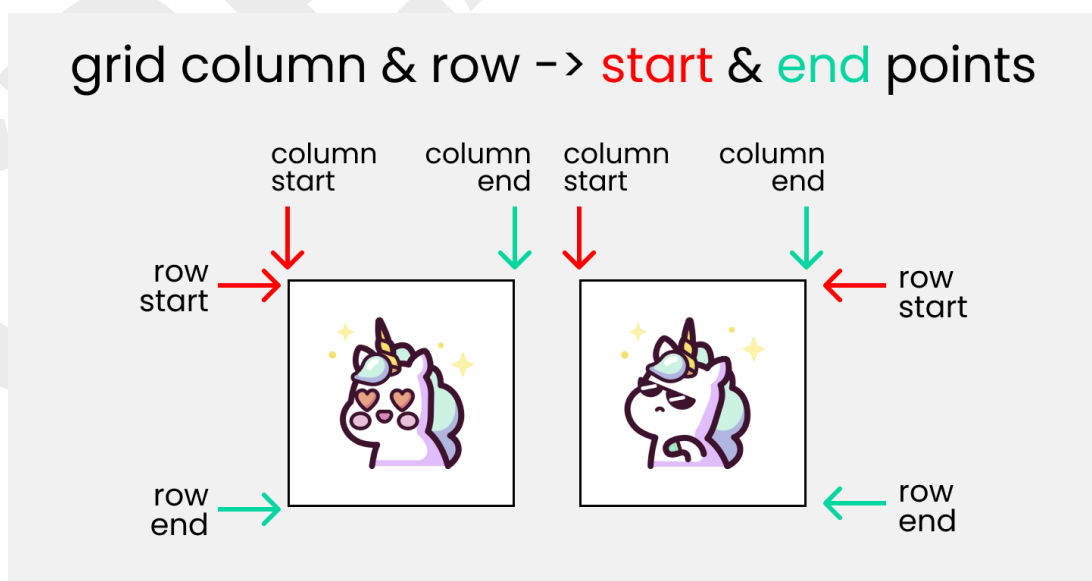


The CSS Grid Scale:

- I made this grid scale to demonstrate the calculations of how rows and columns are joined together. We can use any 1 of the 2 types of measurement:
 - The digit [1,2,3,4, etc...]
 - The span keyword



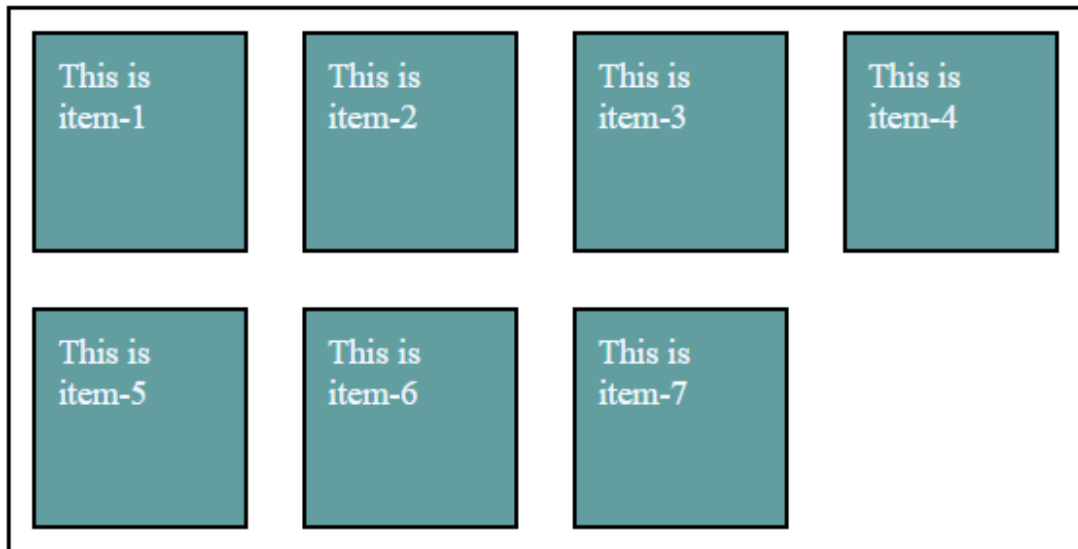
- The illustration below  shows the start and end points of rows and columns of a single cell.



The grid-column: start/end properties :

- You use these two properties to join multiple COLUMNS together. It is a shorthand of 2 properties:
 - **grid-column-start**
 - **grid-column-end**

grid-template-columns: repeat(4, auto);

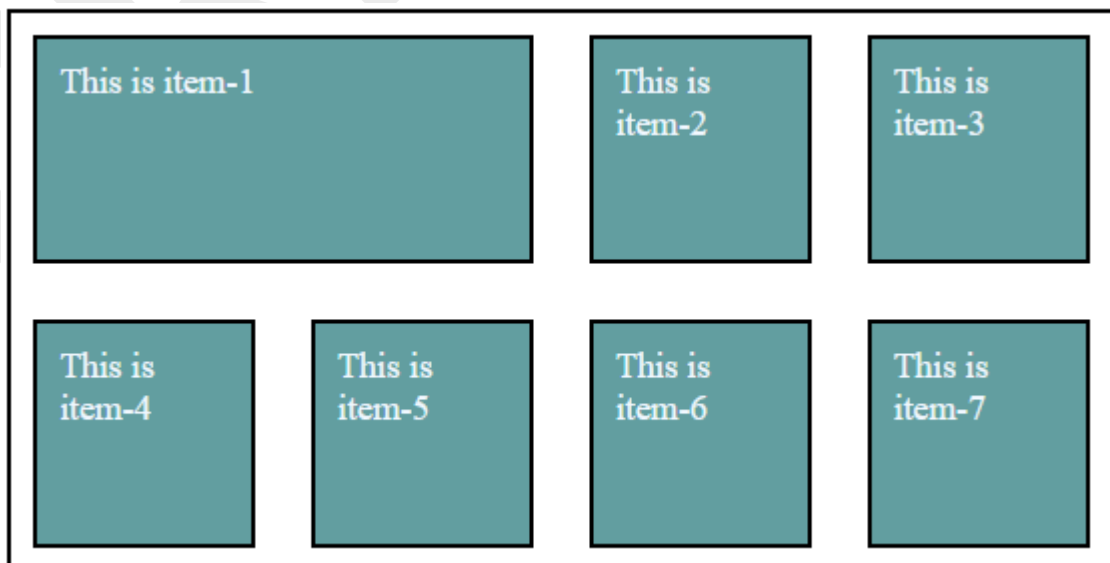


grid-columns-start: 1;

grid-columns-end: 2;

OR

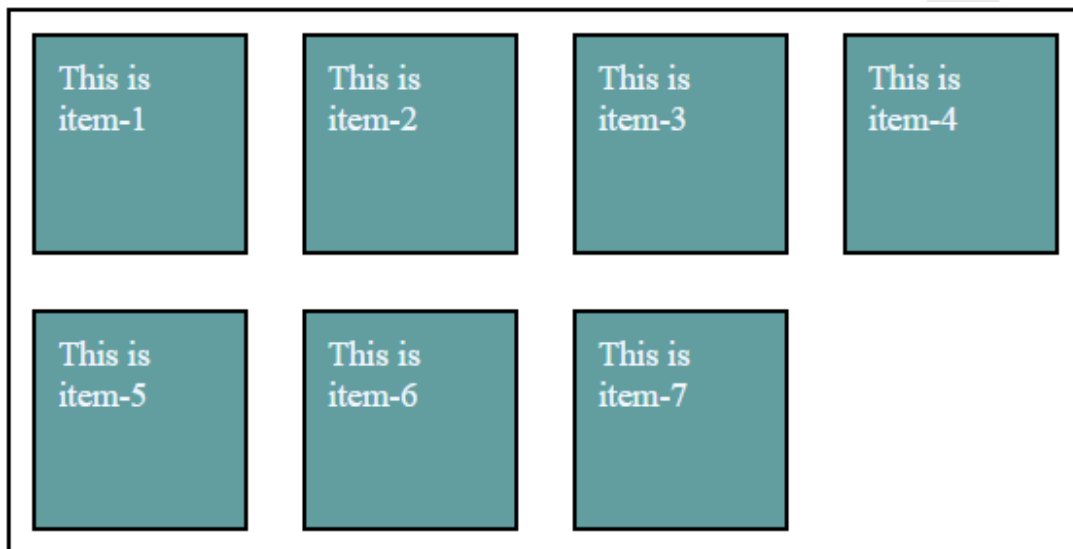
grid-column: 1 / span 2;



The grid-row: start/end properties :

- You use these two properties to join multiple ROWS together. It is shorthand of 2 properties:
 - **grid-row-start**
 - **grid-row-end**

 **grid-template-columns: repeat(4, auto);**

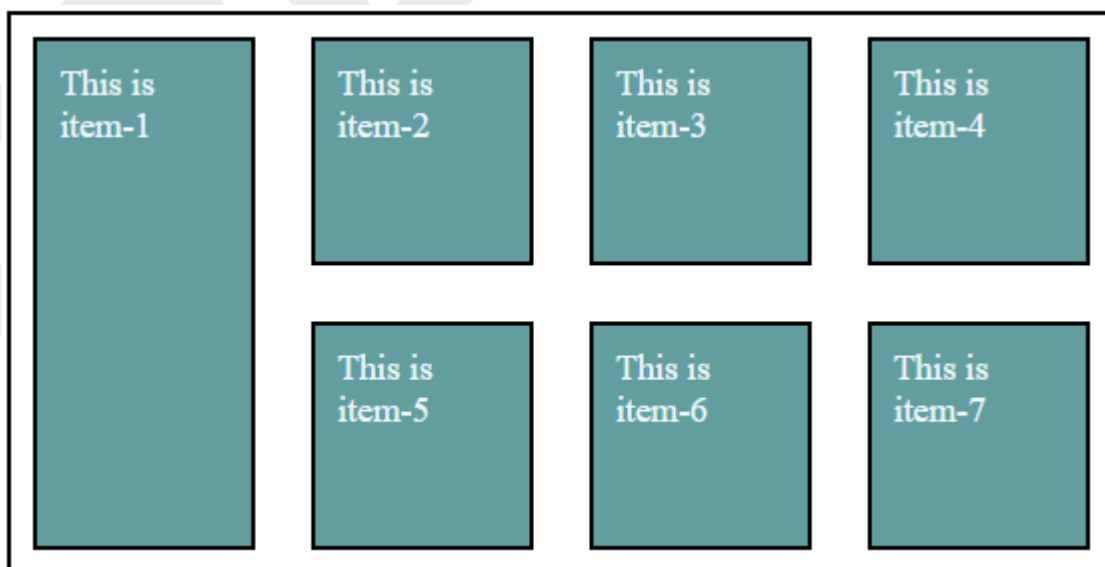


 **grid-row-start: 1;**

 **grid-row-end: 2;**

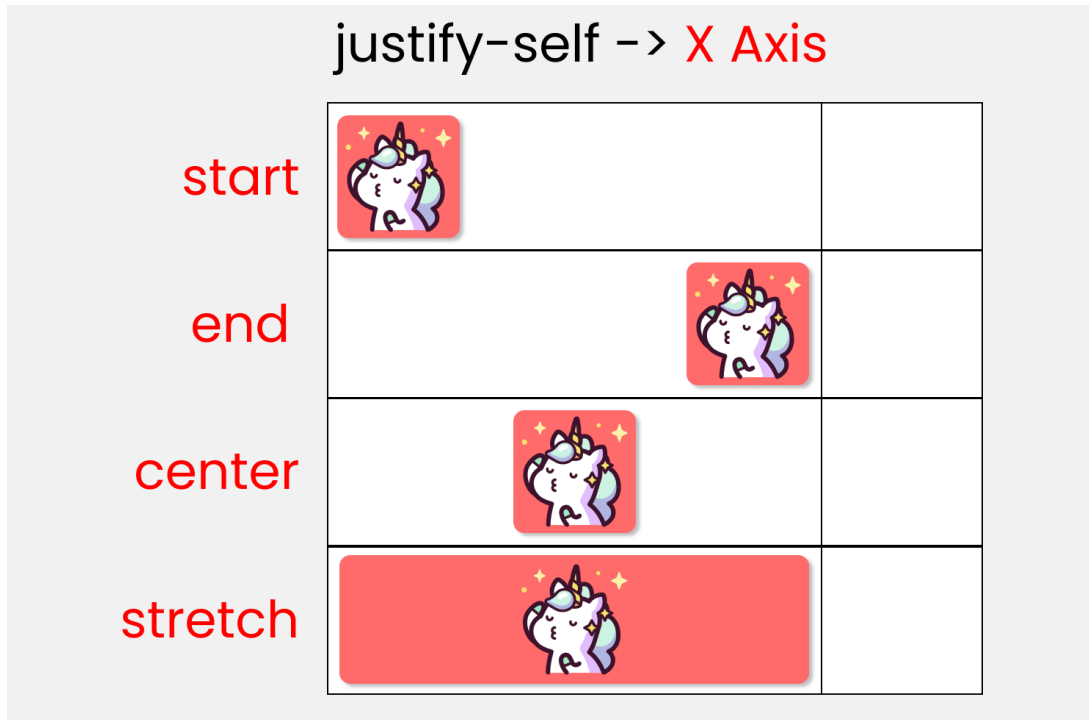
 **OR**

 **grid-row: 1 / span 2;**



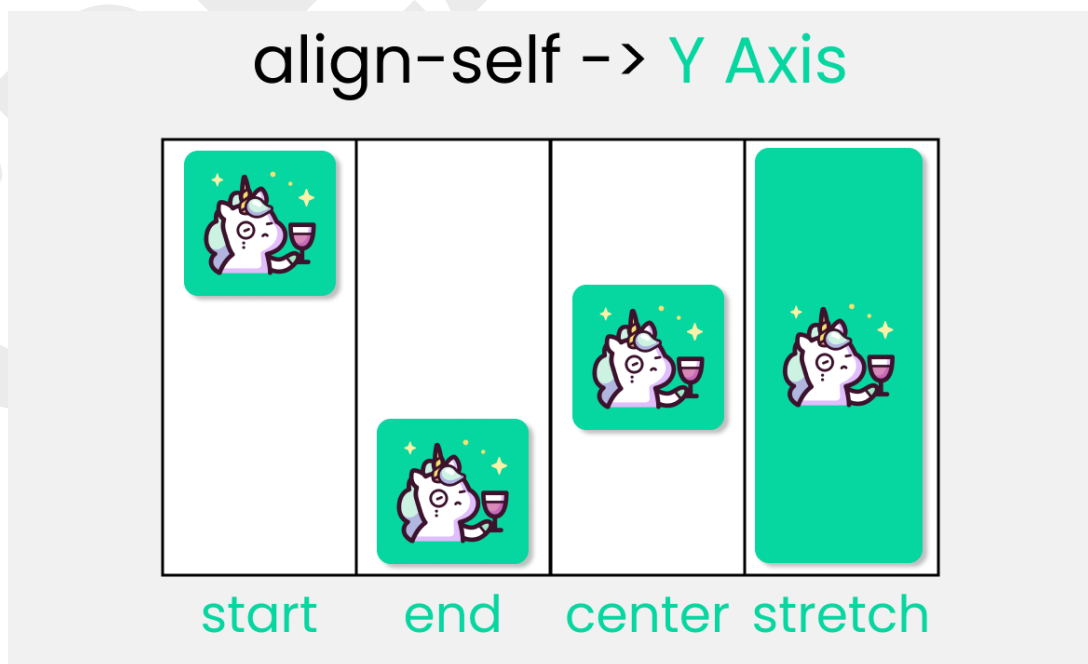
The justify-self property:

- You use this property to position 1 individual grid-item (child) inside a grid container along the X-Axis [Main Axis]. The 4 values are 



The align-self property:

- You use this property to position 1 individual grid-item (child) inside a grid container along the Y-Axis [Cross Axis]. The 4 values are 



Shorthand for CSS Grid Properties

place-content:

- This is the shorthand of 2 properties:
 - **align-content**
 - **justify-content**

place-content : align-content / justify-content

↑
Y-axis

↑
X-axis

place-items:

- This is the shorthand of 2 properties:
 - **align-items**
 - **justify-items**

place-items : align-items / justify-items

↑
Y-axis

↑
X-axis

place-self:

- This is the shorthand of 2 properties:
 - **align-self**
 - **justify-self**

place-self : align-self / justify-self

↑
Y-axis

↑
X-axis

grid-template:

- This is the shorthand of 2 properties:
 - **grid-template-rows**
 - **grid-template-columns**

```
grid-template : grid-template-rows / grid-template-columns
```

```
grid-template : 100px 100px / 200px 200px;
```

gap/grid-gap:

- This is the shorthand of 2 properties:
 - **row-gap**
 - **column-gap**

```
gap : row-gap column-gap;
```